Digital Banking Risk Classification

Theory and Project Report

1. Project Objective

The objective of this project is to build a machine learning classification model that identifies digital banking records as low-risk or high-risk. The target column is risk_status, where 0 represents Low Risk and 1 represents High Risk. The project evaluates how well models detect high-risk cases, especially because high-risk records are less common.

2. Dataset Overview

The project uses Risk_Alert_Classifier_Dataset_4600.csv. The dataset initially contains 4,600 records and 19 columns. The target variable is risk_status. The data is inspected for data types, missing values, duplicate rows, and class distribution before modelling.

3. Data Preparation

- Separate independent variables (X) from the target variable (y).
- Use a stratified train-test split to preserve class proportions in both sets.
- Convert categorical variables to numeric columns using one-hot encoding.
- Handle missing feature values with K-Nearest Neighbors (KNN) imputation. Fit the imputer on training data and apply it to test data.
- Apply sampling techniques only to the training data. Keep the test set unchanged to avoid data leakage.

4. Machine Learning Models

Logistic Regression is used as the baseline classifier. Decision Tree and Random Forest classifiers are also trained. Logistic Regression estimates class probabilities using a logistic function. A Decision Tree learns feature-based decision rules. Random Forest combines multiple decision trees to improve stability and generalization.

5. Class Imbalance and Sampling

The dataset is imbalanced because low-risk records are more common than high-risk records. The training data can be balanced using Random Under-Sampling, Random Over-Sampling, SMOTE, and ADASYN. Under-sampling reduces the majority class, while over-sampling increases the minority class. SMOTE and ADASYN generate synthetic minority-class examples. The test set is not resampled.

6. Evaluation Metrics

- **Accuracy:** The proportion of all predictions that are correct.
- **Precision:** Of the records predicted as high-risk, the proportion that are truly high-risk.
- **Recall (Sensitivity / TPR):** Of the actual high-risk records, the proportion correctly detected.
- **F1 Score:** The harmonic mean of precision and recall.
- **ROC-AUC:** Measures how well the model separates the two classes across thresholds.
- **Confusion Matrix:** Summarizes true negatives, false positives, false negatives, and true positives.

7. Type I and Type II Errors

A Type I error (False Positive) occurs when a low-risk record is incorrectly flagged as high-risk. A Type II error (False Negative) occurs when an actual high-risk record is predicted as low-risk. Type II errors can be particularly important in banking risk screening because a risky case may go undetected. False positives can also create unnecessary reviews.

8. Hyperparameter Tuning and Model Selection

RandomizedSearchCV searches selected Random Forest hyperparameters using cross-validation. Recall can be used as the tuning score because detecting high-risk records is an important objective. Final model selection should consider recall alongside precision, F1 score, ROC-AUC, and the confusion matrix. The best model must be chosen from actual evaluation results.

9. Conclusion

This project demonstrates a workflow for digital banking risk classification, including data exploration, preprocessing, class-imbalance handling, model training, tuning, and evaluation. The model comparison helps identify an appropriate classifier for the dataset. In practical use, the decision threshold and acceptable balance between missed high-risk cases and false alerts should be selected according to banking requirements.